Quiz 5

Question 1 (2 pts)

When a random card is drawn from a standard 52-card deck, the probability that it is red and not a face-card is

- [ ] 5/13
- [ ] 9/26
- [ ] 6/13
- [ ] 10/13

Question 2 (2 pts)

An urn contains five balls numbered 1, 2, 3, 4, and 5. If a person selects a set of two balls at random, what is the probability that value of the sum of two numbers on the balls is bigger than 5 ?

- [ ] 3/5
- [ ] 1/2
- [ ] 2/5
- [ ] 7/10

Question 3 (2 pts)

Which of the following sequences is the solution of the recurrence relation $a_n = \frac{a_{n-1}}{2a_{n-1}-1},\ n = 1, 2, 3, \cdots$

A) $a_n = 2^n - 1$

B) $a_n = 3n + 5$

C) $a_n = \begin{cases} 2, & n = 2k \\ \frac{2}{3}, & n = 2k+1 \end{cases}$

D) $a_n = n^2 + n + 5$

- [ ] B
- [ ] D
- [ ] C
- [ ] A

Question 4 (2 pts)

A person puts an initial deposit of 200 millions VND to his bank account earning interest at a rate of 6% per year compounded monthly and each month he adds an additional 5 millions VND to the account.

For each nonnegative integer n, let A_n be the amount (in millions VND) in the account at the end of n months.
Which of the following is the recurrence relation relating A_n to A_{n-1}.

A) $A_n = 1.06A_{n-1} + 5$

B) $A_n = 200A_{n-1} + 5$

C) $A_n = 1.005^{12}A_{n-1}$

D) $A_n = 1.005A_{n-1} + 5$

- [ ] C
- [ ] B
- [ ] A
- [ ] D

Question 5 (2 pts)

Let a_n be a sequence satisfies the recurrence relation
$a_n = 4a_{n-1} - 4a_{n-2}$, for $n \geq 2$ and $a_0 = 1, a_1 = 3$.

What is the value of a_{20}? (**fill in the exact number in decimal notation**)